Executive Summary

- Overview of the project:
- - Analyzed U.S. wildfire data
- - Tools: Python, SQL, Folium, Plotly Dash
- - Key findings: Western U.S. shows increased fire activity

Introduction

- Wildfires cause extensive damage annually.
- Dataset: Historical_Wildfires.csv
- Goals:
- - Visualize trends
- - Identify high-risk zones
- - Predict wildfire risk

Data Collection & Wrangling

- Data Source: IBM Dataset
- Cleaning Steps:
- - Removed nulls
- - Parsed dates
- - Converted coordinates

EDA & Visual Analytics Methodology

- Steps:
- - Trend and distribution analysis
- - Group by year/state/cause
- - Tools: Pandas, Seaborn, Plotly

Predictive Analysis Methodology

- Objective: Classify fire severity
- Model: Random Forest
- Features: State, Month, Cause
- Train/test split, accuracy measured

EDA Visualization Results 1

- Line chart: Fires per year
- Bar chart: Fires per state

EDA Visualization Results 2

- Histogram: Fire size distribution
- Pie chart: Fire causes

EDA Visualization Results 3

- Box plot: Fire durations
- Heatmap: Feature correlations

EDA SQL Results 1

- Top states by number of fires
- Monthly fire counts

EDA SQL Results 2

- Average duration by cause
- Total area burned per state

EDA SQL Results 3

- Fire causes by region
- YOY wildfire growth

Interactive Map with Folium

- Map showing fire start points
- - Colored by size
- - Clustering enabled

Plotly Dash Dashboard

- Dashboard Features:
- - Filters: Year, State, Cause
- - KPIs: Total fires, Avg size
- - Interactive charts

Predictive Analysis Results

- Accuracy: 85%
- Confusion Matrix, ROC Curve
- Feature Importance chart

Conclusion

- Wildfires rising in western U.S.
- Focus on dry months
- Future: Add weather data

Creativity & Innovation

- Custom visuals, transitions
- Insight: Largest fires peak in October